Summary Notes:
CI/CD on GCP Google Cloud Platform

Overview

Here is an overview of the topics covered in this session.

- Introduction to Continuous Integration (CI)
    - Overview of CI Principles
    - Benefits and Drawbacks of Traditional SDLC
- Phases of Continuous Integration
    - Unit Testing
    - Compilation and Packaging
    - Static Code Analysis
- CI/CD Tools on GCP
    - Cloud Build
    - Cloud Source Repositories
    - Artifact Registry
- Cloud Deploy
    - Introduction to Cloud Deploy
    - Key Features and Benefits
- Testing in CI/CD Pipelines
    - Importance and Types of Automated Testing
- Monitoring and Logging in CI/CD
    - Key Tools and Benefits
- Automating Rollbacks and Approvals
    - Key Concepts and Benefits

Session 1:

Introduction to Continuous Integration (CI)

- **Core Concept**: Continuous Integration automates code integration and testing through a shared repository
- **Key Elements**:
    - Centralised code repository management
    - Automated builds and testing cycles
    - Developer feedback mechanisms
- **Value Proposition**:
    - Enhanced code reliability

    - Quick development cycles
    - Deployment-ready codebase

2. CI Pipeline Components

- **Testing Framework**:
    - Validates code functionality at component level
    - Popular Tools: pytest, JUnit, NUnit
- **Build Management**:
    - Handles code compilation and artifact creation
    - Key Tools: Maven, Gradle, Ant
- **Code Quality Analysis**:
    - Performs automated code review
    - Core Techniques:
        - **Security Scanning**: Identifies code vulnerabilities
        - **Flow Analysis**: Examines data pathways
    - Platforms: SonarQube, Checkmarx

3. GCP CI/CD Solutions

- **Build Automation**:
    - Cloud Build for pipeline orchestration
    - Supports multiple build technologies
- **Code Management**:
    - Cloud Source Repositories for version control
- **Artifact Storage**:
    - Artifact Registry for build output management

Session 2:

1. Cloud Deploy Overview

- **Core Function**: Automated deployment management
- **Primary Features**:
    - Environment-specific deployment pipelines
    - Version management and rollback
    - Compliance-focused approval systems

2. Pipeline Testing Strategy

- **Strategic Value**:
    - Proactive defect identification
    - Continuous developer insights

- **Test Categories**:
    - Component-level unit testing
    - System integration validation
    - Complete application testing

3. Observability Integration

- **Essential Tools**:
    - **Cloud Monitoring**: Performance tracking
    - **Cloud Logging**: Debug information capture
- **Strategic Benefits**:
    - Enhanced pipeline transparency
    - Real-time issue detection

4. Deployment Safety Measures

- **Key Mechanisms**:
    - Production deployment checkpoints
    - Automated version recovery
- **Operational Benefits**:
    - Quality assurance enforcement
    - Minimised incident impact time